ES-FA / BTP Presentation Report

Event-Driven SNN FPGA Accelerator

Local smoke-run summary

May 5, 2026

Current status: software pipeline completed locally. Hardware/Vivado execution was skipped in the latest smoke run. KV260 board measurement is the main pending step.

1. Project Idea

ES-FA studies an event-driven FPGA execution path for spiking neural networks. The basic idea is to use spike sparsity to reduce unnecessary memory access and compute work. Instead of reporting only model accuracy, the project also tracks hardware-facing metrics such as memory access, energy proxy, latency proxy, and hardware validation outputs.

2. What Runs Now

- Baseline LIF SNN training runs locally.
- Hardware-aware and adaptive experiments run locally.
- Analysis generates ranking tables, evidence comparisons, and plots.
- CLI demo works for control, math, and SNN-style classification queries.
- Output sections are generated under `output/`.
- Hardware RTL and validation scripts exist, but were skipped in this smoke run.

3. Current Architecture

Layer	Current role
`p1_training/`	SNN model, training loop, baseline run, INT8/spike exports
`p2_hardware_model/`	spike, memory, energy, and latency proxy estimators
`experiments/`	hardware-aware loss and adaptive dataflow experiments
`analysis/`	ranking table, plots, evidence reports
`hardware/`	Verilog router, scheduler, event queue, memories, LIF PE
`hardware_validation/`	xsim and Vivado flow for KV260-oriented validation
`deployment/`	CLI and adaptive execution policy
`output/`	report, paper, and presentation-ready artifacts

4. Latest Smoke-Run Result

Run	Accuracy	Sparsity	Energy proxy	Score
`exp1_hardware_aware_loss`	0.9570	0.5648	4,592,863.17	0.7699
`baseline_paper1`	0.9570	0.5648	22,604,295.76	0.7387
`exp5_dataflow_adaptation`	0.9570	0.5648	45,016,894.73	0.6699

Main point

The best current software run is `exp1_hardware_aware_loss`. It keeps 95.70% accuracy and gives a 79.68% lower energy proxy compared with the dense baseline estimate. This is a software proxy result, not measured board power.

5. Demo Commands

```
.\scripts\run_esfa_pipeline.ps1 -PythonExe .\.venv312\Scripts\python.exe `
  -Mode smoke -SkipVivado -SkipHardware

python deployment/cli_interface/main.py --query "hi"
python deployment/cli_interface/main.py --query "strike rate 167.55 balls 39"
python deployment/cli_interface/main.py --query "classify results/runtime/sample_signal.
    bin"
```

6. Hardware/Vivado Status

The current RTL and validation stack are present, but the latest completed run skipped hardware using `-SkipVivado -SkipHardware`. Existing local hardware evidence rows are still available in the results folder, but the fresh presentation result should be described as a software smoke run. Final hardware claims should wait for:

- xsim-only validation refresh;
- full Vivado synthesis/implementation refresh;
- physical KV260 board measurements.

7. What To Say In Presentation

> "The software and analysis side is running end-to-end locally. The best hardware-aware run preserves 95.70% validation accuracy and reduces the energy proxy by 79.68% compared with the dense baseline estimate. The RTL and KV260-oriented validation scripts are in the repo, but fresh Vivado execution and physical KV260 board measurements are the remaining hardware tasks."

8. Next Steps

1. Run xsim validation without the `-SkipHardware` flag.
2. Run Vivado validation when build time is available.

3. Boot KV260 and collect physical board measurements.
4. Compare board measurements with software estimators.
5. Tune estimator constants and rerun the report.